Experiment 2(a): Sort an Array of Data in Ascending Order Using 8085

AIM:

To write an assembly language program to sort an array of data in ascending order. The array is stored in memory starting from 4200H, and the first element gives the number of elements in the array.

APPARATUS REQUIRED:

1. 8085 Microprocessor kit – 1
2. Power card – 1
3. Keyboard – 1

ALGORITHM:

1. Load the number of elements into register B.
2. Decrement B to get the number of passes.
3. Initialize the HL pair to point to the array.
4. Load the first element into the accumulator.
5. Compare it with the next element.
6. Swap the elements if they are in the wrong order.
7. Repeat the comparisons until the array is sorted.
8. Stop the program.

PROGRAM:

```
LXI H,4200H
MOV B,M
DCR B

OUTER: MOV C,B
LXI H,4201H

INNER: MOV A,M
INX H
CMP M
JC SKIP
JZ SKIP
MOV D,M
```

MOV M,A

DCX H

MOV M,D

INX H

SKIP: DCR C

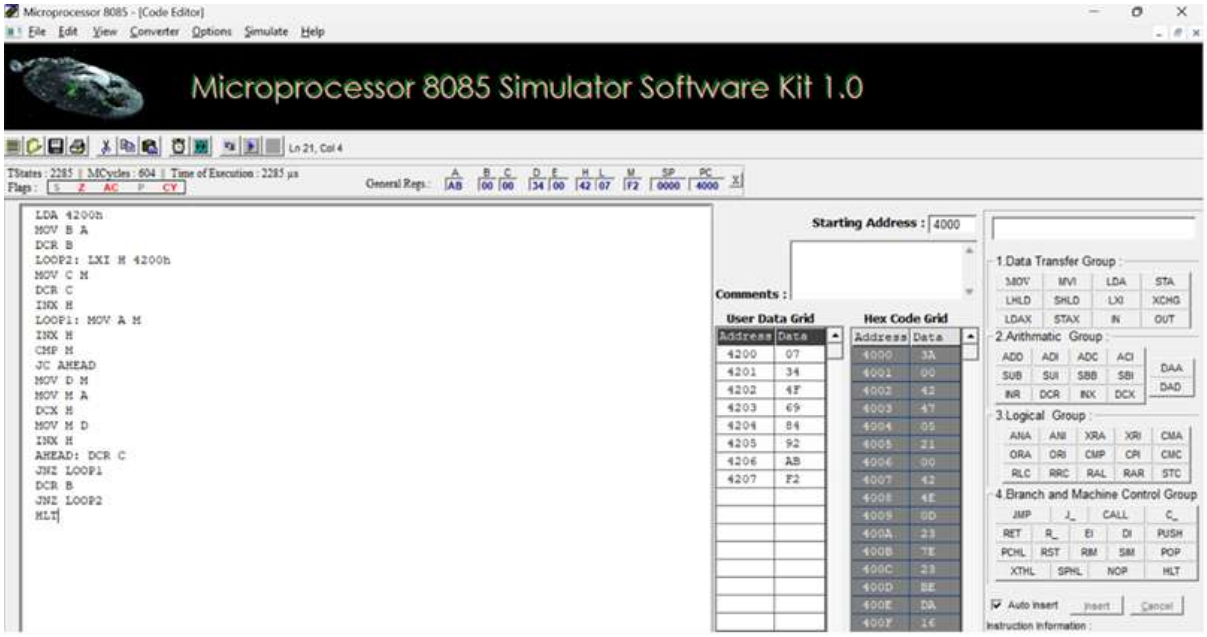
JNZ INNER

DCR B

JNZ OUTER

HLT

OUTPUT:



RESULT:

Thus, an assembly language program to sort an array of data in ascending order using the 8085 microprocessor was written and executed.